LILACS lab

Literacy, Language, & Content Learning in Education Sciences

At the LILACS lab (Literacy, Language, and Content learning in Education Sciences), directed by Dr. HyeJin Hwang at UMN, we are driven to understand the intricate pathways through which young students build comprehension and knowledge. We seek to identify evidence-based strategies that better support student learning across all content areas.

Learn more about the lab | Meet our team | Explore research projects | See publications | Lab news | Contact information

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What We Do:

- The Literacy, Language, and Content Learning in Education Sciences (LILACS) Lab investigates how K–12 students learn through engagement with various types of texts—across genres and modalities—and leverages these research findings to design and evaluate effective instructional strategies and educational technologies.
- Our current research particularly focuses on multilingual elementary students' language and reading comprehension as well as their science and social studies knowledge development.
- We collaborate closely with teachers and students in schools and are dedicated to bridging the gap between research and educational practice and policy.



2026 Fall

Guiding Principles for Our Work:

- We view reading comprehension as both processes and products. We investigate not only the final outcome of what students comprehend, but also the active, cognitive strategies they employ while engaging with various texts.
- We believe reading is a fundamental vehicle for learning and growth. For example, by interacting with text, students can learn about the natural and social worlds and build epistemological virtues (e.g., intellectual humility).
- We hold an asset-oriented view of multilingualism. We view students' home languages and cultural backgrounds as powerful resources that enhance learning.
- We are committed to interdisciplinary research. Housed in educational psychology and special education, we actively bridge insights from various disciplines such as cognitive psychology and learning sciences

Get to know the researchers behind the work.

Our team

Meet our people

Meet our team

Our Team

Lab Director



LAB DIRECTOR

HyeJin Hwang, Ph.D.

Assistant Professor · University of Minnesota

"I study how students learn from diverse texts because I believe reading is the fundamental means for learning."

Meet HyeJin ↗

Lab Coordinator & Ph.D. student

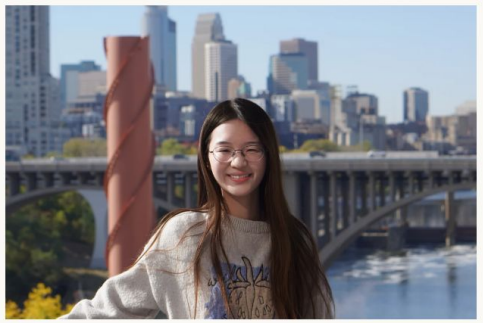


LAB COORDINATOR · PH.D. STUDENT

Yoo Jeong Son ↗

Our Team

Ph.D. students



PH.D. STUDENT

Yulu Sun ↗

My research connects psychological research and educational practice to improve children's reading and learning, particularly among children with developmental dyslexia.

Master's students



MASTER'S STUDENT

Jianjian Wu ↗

I study reading comprehension and literacy teacher education because reading can support intellectual growth and personal flourishing.



MASTER'S STUDENT

Kiera Rosenberg ↗

I study the intersection of math and language because I believe cognition in these fields is more related than we think.

Collaborating labs at UMN

- R+L lab directed by Dr. Panayiota (Pani) Kendeou
- McMaster lab directed by Dr. Kristen McMaster
- Inference Galaxy lab coordinated by Susan Slater

Alumni & Collaborators

- Alumni and Collaborators
  - Seohyeon Choi, Ph.D. – Assistant Professor of Child Psychology & Education at Sungkyunkwan University in South Korea)
  - Christopher Blommel – Teacher at Japan Exchange and Teaching Program)
- Collaborators
  - Yewon Kang – Ph.D. candidate at UMN
  - Crystal Wise, Ph.D. – Assistant Professor at the Dpt of Curriculum & Instruction at UMN
  - Laura Allen, Ph.D. – Associate Professor at the Dpt of Ed Psy at UMN
  - Sonia Cabell, Ph.D. at FSU
  - Liz Hadley, Ph.D. at U of Florida
  - Jackie Relyea, Ph.D. at NCSU
  - Miranda Fitzerald, Ph.D. at NCSU
  - Courtney Hattan, Ph.D. at UNC
  - Sungyoon Lee, Ph.D. at Middle Tennessee University
  - Ye Shen, Ph.D. at Johns Hopkins University

Note: I'm still working on this…



BEYOND THE LAB

I enjoy reading novels and watching movies. My favorite movie directors include Bong Junho, Jang Hangjun, and Yoon Ga-eun.

LAB DIRECTOR · ASSISTANT PROFESSOR

HyeJin Hwang

Ph.D., University of Michigan

I study how students learn from diverse texts because I believe reading is the fundamental means for learning.

HyeJin is an Assistant Professor in the Department of Educational Psychology (Psychological Foundations and Special Education) at the University of Minnesota. She earned her Ph.D. at the University of Michigan and previously taught ESL/EFL in Korea.

Research interests

- Developmental processes of reading comprehension and content knowledge in science and social studies
- Instructional approaches and educational technologies that support reading comprehension and content learning
- Multilingual students in the elementary years

Current work

See the research project page ↗

University profile ↗

Teaching

- EPSY 5001 Learning, Cognition, Assessment (undergraduate course for preservice teachers)
- EPSY 5216 Introduction to Research in Educational Psychology and Human Development (research methods course for graduate students)
- EPSY 8116 Reading for Meaning: Cognitive Processes in the Comprehension of Texts (doctoral seminar course)

Advising and mentoring

- Personal statement about advising and mentoring: Advising and mentoring are critical components of my work. I view advising and mentoring as opportunities to grow together with advisees/mentees as scholars and human beings. I recognize my power and privilege that come from being an able-bodied, cis-gendered, Western-educated, English-speaking person, as well as from being a faculty member at a university.
- To maximize your achievement and our collaborative learning, I have the following commitments to you as my student:
    - Cultivating a safe lab environment that values intellectual humility
    - Maintaining honest and respectful communication in a timely manner
    - Providing explicit guidelines and individualized scaffolding to support you in achieving your goals and program milestones
    - Unpacking hidden curricula in academia and research
- To maximize your achievement and our collaborative learning, I expect following things from you as my student:
    - Aligned Research Interests: A commitment to research inquiry that aligns closely with the lab's core focus areas
    - Intellectual humility – It helps us continuously learn from various perspectives and different types of feedback.
    - Honest and respectful communication in a timely manner
    - Proactive planning: Expect that we will start strategic goal setting and step-by-step planning early in your program and continue this practice at the beginning of each semester.



BEYOND THE LAB

I enjoy spending time in nature and going camping.

LAB COORDINATOR · PH.D. STUDENT

YooJeong Son

I study K–12 students' reading comprehension across print, online, and AI-mediated environments.

YooJeong was an elementary school teacher. Building on her classroom experience, she is interested in translating research findings into educational practices in K–12 settings. Her work focuses on designing interventions and examining the psychological factors that influence comprehension.

Research interests

- Content-area literacy to build robust knowledge
- Online evaluation of multimodal and AI-generated information
- Child–AI interaction during reading and writing

Current work

- RAISE
- Lateral reading
- ROSCO
- CLAP
- Digital Reading Comprehension (DRC)

Explore lab research ↗

- Knowledge revision: Read-Aloud Instruction for SciencE Learning (RAISE Learning)
- Lateral reading intervention
- Standards of Coherence: Measurement development studies
- Supporting AI literacy within content-rich literacy instruction: Content-rich Literacy and AI literacy via Project-based learning (CLAP).
- Systematic review on digital reading comprehension

← All people



BEYOND THE LAB

I enjoy hiking, reading novels, and watching funny TV shows.

PH.D. STUDENT

Yulu Sun

My research connects psychological research and educational practice to improve children's reading and learning, particularly among children with developmental dyslexia.

Yulu investigates the cognitive mechanisms underlying reading and learning in typical and atypical childhood development. She also examines how literacy interactions with educational stakeholders shape children's language and cognitive growth.

Research interests

- Cognitive mechanisms of reading and learning
- Literacy interactions and children's language and cognitive development
- Evidence-based, AI-enabled agents for literacy assessment and intervention at school and at home

Current work

- Supporting inference-making using educational technology: Inference Galaxy
- Systematic review on digital reading comprehension

Explore lab research ↗



BEYOND THE LAB

I enjoy swimming, hiking, traveling, and watching movies and TV shows. I have a cat named Mimi.

MASTER'S STUDENT

Jianjian Wu

I study reading comprehension and literacy teacher education because reading can support intellectual growth and personal flourishing.

Jianjian is interested in teachers' roles in children's academic and social–emotional development. Her work includes support for an international project involving elementary students in South Korea, including survey data entry and practical reports for teachers and parents.

Research interests

- How teachers support K–12 multilingual students' reading motivation across languages
- The emotional and motivational aspects of learning and teaching
- Interactive reading interventions for young readers who experience reading difficulties

Current work

- ROSCO 2
- Inference Galaxy
- WRITE
- Systematic review of reading motivation among multilingual children and adolescents

- Supporting inference-making using educational technology: Inference Galaxy
- Standards of Coherence: Measurement development studies
- Systematic review of reading motivation among multilingual children and adolescents

Languages & research tools

Languages: Chinese (native), English (fluent). Research tools: Python, R, and Mplus.

Explore lab research ↗

OUR ACADEMIC HOME

Department of Educational Psychology
University of Minnesota
Minneapolis, Minnesota

University faculty profile ↗

LET'S CONNECT

Research & graduate mentoring

hwang305@umn.edu ↗

Lab director · HyeJin Hwang, Ph.D.



BEYOND THE LAB

I have two cats who are very spoiled. My hobbies include playing video games, cooking, crocheting, and reading novels.

MASTER'S STUDENT

Kiera Rosenberg

I study the intersection of math and language because I believe cognition in these fields is more related than we think.

Kiera examines numerical and linguistic meaning-making, cross-cultural differences in cognition, and the role of reading comprehension in mathematics.

Research interests

- Numerical and linguistic meaning-making
- Cross-cultural differences in numerical and linguistic cognition
- The role of reading comprehension in mathematics

Current work

- Automated analysis of non-linear revision behaviors during timed essay writing
- Systematic review on digital reading comprehension
- Understanding chronic absenteeism in collaboration with school district leaders

Explore lab research ↗

Two Lines of Inquiry

1. We investigate developmental processes of reading comprehension and content Knowledge.
2. We investigate instructional approaches to support reading comprehension and content knowledge, building on the first line of research.

We study both printed and online/digital texts. Our research often focuses on elementary years and multilingual students.

Current Research Projects

Developmental Processes

1. Standards of coherence

Standards of coherence are the implicit and explicit criteria readers use to evaluate whether their understanding of the text is sufficient to meet their reading goals. We develop and validate measures for standards of coherence and examine individual and developmental differences in them (e.g., Hwang et al., 2025 August).

2. Heterogeneity in literacy and language development

We investigate heterogeneity in literacy (e.g., word recognition) and language development (e.g., vocabulary) and their contribution to reading comprehension and content learning (e.g., Hwang et al., 2026; Shen & Hwang, 2026). This work includes review studies on digital reading comprehension.

Instructional Approaches

3. Lateral reading

Students increasingly interact with online texts to learn. As such, it is critical to support online reading strategies. Lateral reading is an online reading strategy to detect misinformation. We design and test an intervention for lateral reading for upper-elementary students (e.g., Son et al., 2026 August).

4. Supporting AI literacy within content-rich literacy instruction

We investigate approaches to supporting AI literacy during content-rich literacy instruction. This work includes Content-rich Literacy and AI literacy via Project-based learning (CLAP).

5. Knowledge revision

Knowledge revision supports knowledge building. We design and examine interventions to support students' knowledge revision during literacy instruction. This includes a project titled Read-Aloud Instruction for SciencE Learning (RAISE Learning; e.g., Hwang et al., 2026).

6. Inference-making

Supporting inference-making is critical as many young students often have difficulties in activating and integrating content knowledge with text. We work with the Inference Galaxy lab to design and test instructional technology in supporting young students' inference-making and content knowledge development.

RESEARCH IN CONVERSATION

Get in touch about our work.

Contact the lab ↗

LILAC LAB

Literacy, Language, and Content Learning Lab

OUR ACADEMIC HOME

Department of Educational Psychology
University of Minnesota
Minneapolis, Minnesota

University faculty profile ↗

LET'S CONNECT

Research & graduate mentoring

hwang305@umn.edu ↗

Lab director · HyeJin Hwang, Ph.D.

We need publications page.

News page – this looks good.

Contact

Have questions about our research or interested in collaboration or joining us? We would love to hear from you.

Email: hwang305@umn.edu

Location: 250 Education Sciences Building, 56 East River Road, Minneapolis, MN 55455

Prospective graduate students

More information on the graduate program in the Department of Educational Psychology can be found in the department website. Dr. Hwang is affiliated with the Psychological Foundations of Education/Learning Technology (primary program) and Special Education (secondary program).

See Dr. Hwang's statements about advising and mentoring